Lecture 16, DAA   Mar 18

A subset system for which greedy algorithm yields the maximum valued feasible subset is called a Matroid $M = (S, \mathcal{I})$ — S: groundset, $\mathcal{I}$: feasible subsets

$$\boxed{\begin{array}{ccc} \text{Matroid} & \Rightarrow & \text{Exchange Property holds} \\ & \nwarrow & \Downarrow \\ & & \text{Rank property holds} \end{array}}$$

— Exchange Property

Given feasible subsets S_K of size K and S_{K+1} of size $K+1$, $\exists\, e \in S_{K+1} - S_K$ s.t. $S_K \cup \{e\}$ is feasible

— Rank Property : For any $A \subseteq S$ the maximal feasible subsets of A have the same cardinality

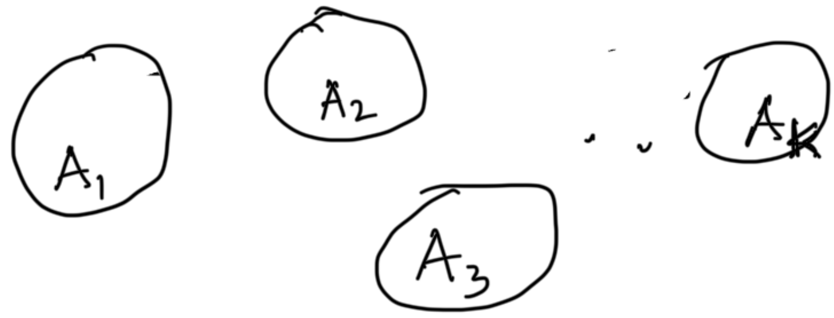
Applications

1. Maximum Spanning Forest (Tree is a special case for connected graphs)

S = set of Edges $\mathcal{I}$: all forests

Use rank property for K connected components in $A \subseteq S$

edges = Induced graph $G_A = (V_A, E_A)$



$|A_i| = n_i$

Maximal # edges $= \sum_{i=1}^{K} (n_i - 1) = \underline{\sum n_i - K}$

invariant for any maximal subset

Maximum v.s. Minimum

Given a minimization problem (on matroid)

1. Redefine $w'(e) = w_{max} - w(e) \geq 0$
   w: original weight function

2. Solve the maximization problem for w', suppose output $= T$

Claim : T is the solution for the minimization problem for the original weight w

$$w(T) = \sum_{e \in T} w'(e) \geq w(T'), \; T' \neq T$$

(maximization soln)

$$\Rightarrow \sum_{e \in T} (w_{max} - w(e)) \geq \sum_{e \in T'} (w_{max} - w(e))$$

$$\Rightarrow |T| \, w_{max} - \sum_{e \in T} w(e) \geq |T'| \, w_{max} - \sum_{e \in T'} w(e)$$

$$\Rightarrow |T| \, w_{max} + \sum_{e \in T'} w(e) \geq |T'| \, w_{max} + \sum_{e \in T} w(e)$$

$$\Rightarrow \forall T' \sum_{e \in T'} w(e) \geq \sum_{e \in T} w(e)$$

from $\boxed{\text{Rank}}$ property

Proof Rank property $\Rightarrow$ Greedy always works

By contradiction

rank in A ← Optimal | $e_1 \; e_2 \; e_3 \; e_4 \quad e_l \quad e_m$

is at least x

Greedy | e_1' e_2'' e_3' e_4' e_ℓ' e_m'

$< \ell$

contradiction of rank(A) $\longrightarrow$ increasing order of weights

Since Optimal > Greedy let ℓ be the smallest index where $w(e_\ell) < w(e_\ell')$

Consider $A = \{$ all elements whose weights $\leq w(e_\ell) \}$ $e_\ell' \notin A$

Maximal subsets should be same from Rank property

So why did greedy choose e_ℓ' ?

Job Scheduling Problem

Example

	J_1	J_2	J_3	J_4
deadlines	4	2	1	2
Penalty	10	15	5	12
Duration	1	1	1	1

T=0

1 2 3 4 5

Schedule as many Jobs to minimize total penalty

Idea: Schedule greedy in decreasing order of penalty

Schedule : set of jobs that can meet the deadline

Maximize the sum of penalty of a "feasible schedule"

all jobs in the subset can be processed within their respective deadlines

Problem 1 : Given a subset S of jobs $\{J_{i_1}, J_{i_2}, J_{i_3} \cdots J_{i_k}\}$
can we determine if S is feasible and if so, produce a feasible schedule

Idea: If we schedule jobs in increasing order of deadlines
and succeed $\rightarrow$ great
fail $\longrightarrow$ doesn't imply no feasible schedule?

Framework of subset system
S = set of all jobs $\mathcal{I}$: all subsets of jobs that can be scheduled without missing deadlines

Question: Does it satisfy the rank property or exchange property?